

← Back3-DES and 4-DESGraded Quiz • 30 min

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Module Objectives

Double-DES and Meet-in-the-Middle Attack

Triple DES

Video: Triple DES3 min

Quiz: 3-DES and 4-DESSubmitted

Advanced Encryption Standard (AES)

🎉 Congratulations! You passed!

Grade received100%

Latest Submission3-DES and 4-DES

To pass 60% or higher

Retake the assignment in 7h 57m

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Review Learning Objectives

1. Triple-DES or 3-DES encryption can be characterized by the following: $C = \text{Enc}(K_3, \text{Dec}(K_2, \text{Enc}(K_1, P)))$. Each keys, K_1, K_2, K_3 are 56-bits-long and are independent to each other (the three-key version). The DES block size is 64 bits.

4 / 4 points

🔒 Submit your assignment

Using the big O notation, which of the following best describe the meet-in-the-middle attacker's storage requirement?

Due Jun 2, 11:59 PM ISTAttempts 3 every 8 hours

Try again

Retake the quiz in 7h 57m

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To Pass 60% or higher

Your grade100%

View Feedback

We keep your highest score

☐ $O(2^{*112})$

☐ $O(168)$

☐ $O(2^{*168})$

☒ $O(2^{*56})$

👍 Like

👎 Dislike

🚩 Report an issue

👉 Correct

This corresponds to the number of outcomes for the shorter direction for the meet-in-the-middle attack.

2. Triple-DES or 3-DES encryption can be characterized by the following: $C = \text{Enc}(K_3, \text{Dec}(K_2, \text{Enc}(K_1, P)))$. Each keys, K_1, K_2, K_3 are 56-bits-long and are independent to each other (the three-key version). The DES block size is 64 bits.

4 / 4 points

Using the big O notation, which of the following best describe the meet-in-the-middle attacker's encryption/decryption computational effort?

☒ $O(2^{*112})$

☐ $O(2^{*128})$

☐ $O(2^{*28})$

☐ $O(2^{*168})$

☐ $O(2^{*56})$

👉 Correct

In one direction the key has 112 bits of entropy while the other direction has 56 bits of entropy.

3. Triple-DES or 3-DES encryption can be characterized by the following: $C = \text{Enc}(K_3, \text{Dec}(K_2, \text{Enc}(K_1, P)))$. Each keys, K_1, K_2, K_3 are 56-bits-long and are independent to each other (the three-key version). The DES block size is 64 bits.

4 / 4 points

The attacker now does not have known plaintext-ciphertext pair that it can use for her cryptanalysis. Using the big O notation, which of the following best describe the attacker's encryption/decryption computational effort?

☒ $O(2^{*168})$

☐ $O(2^{*112})$

☐ $O(2^{*56})$

☐ $O(2^{*128})$

👉 Correct

The attacker cannot perform meet-in-the-middle attack, which requires known plaintext.

4. Quadruple-DES or 4-DES encryption can be characterized by the following: $C = \text{Enc}(K_4, \text{Enc}(K_3, \text{Enc}(K_2, \text{Enc}(K_1, P))))$. Each keys, K_1, K_2, K_3, K_4 , are 56-bits-long and are independent to each other. The DES block size is 64 bits.

4 / 4 points

Using the big O notation, which of the following best describe the meet-in-the-middle attacker's storage requirement?

☐ $O(2^{*56})$

☒ $O(2^{*112})$

☐ $O(2^{*224})$

☐ $O(64^{*2^{*56}})$

☐ $O(2^{*128})$

👉 Correct

Meet-in-the-middle attacker computes from both directions to find a match on the intermediate block values, which is of 64 bits long. The key size for each direction is 112 bits long, which produces 2^{*112} trials/options for the attacker to try.

5. Quadruple-DES or 4-DES encryption can be characterized by the following: $C = \text{Enc}(K_4, \text{Enc}(K_3, \text{Enc}(K_2, \text{Enc}(K_1, P))))$. Each keys, K_1, K_2, K_3, K_4 , are 56-bits-long and are independent to each other. The DES block size is 64 bits.

4 / 4 points

Using the big O notation, which of the following best describe the meet-in-the-middle attacker's encryption/decryption computational effort?

☐ $O(2^{*224})$

☐ $O(2^{*58})$

☐ $O(2^{*56})$

☐ $O(2^{*128})$

☒ $O(2^{*112})$

👉 Correct

Each of the directions for the meet-in-the-middle attack has 112 bits of entropy, which corresponds to that of K_1 - K_2 or K_3 - K_4 .

6. Quadruple-DES or 4-DES encryption can be characterized by the following: $C = \text{Enc}(K_4, \text{Enc}(K_3, \text{Enc}(K_2, \text{Enc}(K_1, P))))$. Each keys, K_1, K_2, K_3, K_4 , are 56-bits-long and are independent to each other. The DES block size is 64 bits.

4 / 4 points

The attacker now does not have known plaintext-ciphertext pair that it can use for her cryptanalysis. Using the big O notation, which of the following best describe the attacker's encryption/decryption computational effort?

☐ $O(2^{*56})$

☐ $O(2^{*58})$

☐ $O(2^{*112})$

☐ $O(2^{*128})$

☒ $O(2^{*224})$

👉 Correct

Without meet-in-the-middle attack, the key entropy is 224 bits.

https://www.coursera.org/learn/symmetric-crypto/exam/5y2gn3-des-and-4-des/attempt/redirectToCover=true

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